L&T EduTech Project Submission

AI-Based Road Surface Defect Monitoring System

Day 2: Road Defect Annotation & Quality Engineering

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2.1 Workspace Setup & Annotation Platform

Day 2 was exclusively dedicated to the most labor-intensive and critical phase of supervised machine learning: structuring raw pixel data into machine-readable ground truth. We utilized Roboflow as our centralized annotation, dataset management, and version control platform. Roboflow allowed simultaneous multi-user collaboration, ensuring the team could divide the massive image corpus efficiently while maintaining annotation consistency.

2.2 Annotation Protocols & Quality Assurance

To prevent the YOLOv8 model from learning incorrect spatial features, we established strict bounding box protocols.

- **Tight Encapsulation:** Bounding boxes had to encapsulate localized defects with less than a 5% pixel margin. Excessive background padding forces the model to associate healthy asphalt with defect features, driving down precision.
- **Segmentation by Severity:** Road cracks often span the entire length of a frame. If a longitudinal crack transitioned from Mild (D10) to Severe (D00) along the fault line, annotators were instructed to split the defect into multiple, distinct bounding boxes rather than drawing one massive box that averages the severity.
- **Edge Case Handling:** Tar sealings, fresh paint, and tire marks share pixel intensity characteristics with deep cracks. These were explicitly left unannotated to force the neural network's convolutional layers to extract deeper, structural gradient features rather than relying on simple color contrast.

2.3 Statistical Review & Addressing Class Imbalance

Following the manual annotation of the core dataset, a statistical health check was executed. The dataset exhibited a heavy long-tail class imbalance. Smooth road frames (D60) and mild cracks (D10) accounted for over 70% of the annotations, while severe potholes (D20) were significantly underrepresented.

Training on this raw distribution would result in a model biased toward predicting "Mild Cracks" simply because it is statistically safer. To rectify this, we utilized targeted under-sampling for the D60 class and deliberate over-sampling for D20 and D00 classes, ensuring a relatively symmetrical distribution across the training batches.

2.4 Synthetic Augmentation Pipeline

To harden the model against the environmental challenges identified on Day 1, we constructed a comprehensive programmatic augmentation pipeline within Roboflow. Augmentations multiply the dataset size by artificially modifying existing images, forcing the model to achieve true generalization.

- **HSV & Brightness Adjustments:** Hue, Saturation, and Value shifts ($\pm$15% brightness, $\pm$10% saturation) were applied. This simulates varying times of day, compensating for the difference between a harsh Indian midday sun and an overcast Norwegian afternoon.
- **Weather Overlays:** We injected synthetic noise, including simulated rain streaks and 10% opacity fog filters. This trains the model to detect edge gradients even through atmospheric interference.
- **Spatial Transforms:** For the drone dataset, random cropping and subtle rotations ($\pm 5^\circ$) were applied. This mimics the inherent instability of UAV gimbals and variations in flight altitude, ensuring scale and rotation invariance.

2.5 Dataset Splitting & Final Export

By the conclusion of Day 2, both the ground-level and aerial datasets were fully annotated, augmented, and reviewed. The data was split into an 80-10-10 ratio: 80% for Training, 10% for Validation (used for hyperparameter tuning during training), and 10% for Testing (held out completely for final unbiased evaluation). The final dataset versions were compiled and exported in the YOLOv8 PyTorch ‘.txt‘ format, ready for immediate ingestion by the training scripts on Day 3.